

Tuning Results

Tuned lattices on Github:
https://github.com/Sievert-L/FCCEE_Project_2026

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FCC-ee LCC V106.2.3 @ Z

Assumed error tolerances

Section	Element	Misalignment			Rotation	Field error (rel.)
		σ_x (μm)	σ_y (μm)	σ_s (μm)	σ_θ (μrad)	
ARC	Dipole	1000	1000	500	1000	$k_0: 1 \times 10^{-3}$
	Quadrupole	50	50	100	50	$k_1: 2 \times 10^{-4}$
	Sextupole	50	50	100	50	$k_2: 2 \times 10^{-4}$
	Girder	150	150	500	150	–
STRAIGHT SECTION	IR Dipole	1000	1000	100	1000	$k_0: 1 \times 10^{-3}$
	TR Dipole	1000	1000	500	1000	$k_0: 1 \times 10^{-3}$
	FD Quadrupole	10	10	100	10	$k_1: 1 \times 10^{-5}$
	FF Quadrupole (excl. FD)	30	30	100	30	$k_1: 1 \times 10^{-4}$
	TR Quadrupole	100	100	100	100	$k_1: 2 \times 10^{-4}$
	FF Sextupole (incl. CS)	30	30	100	30	$k_2: 1 \times 10^{-4}$
I	BPM	10	10	–	10	–

Optics before correction

Observable	LCC Z-pole (V106.2.3)
rms orbit H / V (μm)	No closed orbit
rms $\Delta D_{x/y}$ (mm)	19.9 / 44.2
rms $\Delta \mu_{x/y}$ [10^{-4} 2π]	98.6 / 334.9
rms $\Delta \beta_{x/y} / \beta_{x/y}$ (%)	14.0 / 34.3
rms $ f_{1001} / f_{1010} $ [10^{-4}]	1696 / 310

Tuning results

Observable	LCC Z-pole (V106.2.3)
failed seeds	7 / 100
rms orbit H / V (μm)	137 / 138
rms H / V orbit corrector strength in ARC (μrad)	11.55 / 11.63
rms H / V orbit corrector strength in IR (μrad)	2.23 / 2.11
rms H / V orbit corrector strength in TR (μrad)	4.66 / 4.67
rms normal / skew optics corrector strength (mT)	0.77 / 0.50
rms $\Delta D_{x/y}$ (mm)	0.66 / 1.55
rms $\Delta\mu_{x/y}$ [10^{-4} 2π]	13.2 / 27.3
rms $\Delta\beta_{x/y}/\beta_{x/y}$ (%)	0.55 / 0.67
rms $ f_{1001} / f_{1010} $ [10^{-4}]	26.4 / 21.8
median ε_x (nm) / ε_y (pm) [from Twiss method]	0.71 / 0.028
transverse tunes Q_x/Q_y	194.16 / 170.20
rms $\Delta D_{x/y}^{(2)}$ (mm)	363 / 635

Corrected lattices with beam-beam @ 75 %

Observable	LCC Z-pole (V106.2.3)
number of lattices	82
rms $\Delta D_{x/y}$ (mm)	1.6 / 9.0
rms $\Delta \mu_{x/y}$ [10^{-4} 2π]	27.1 / 1216
rms $\Delta \beta_{x/y} / \beta_{x/y}$ (%)	2.12 / 42.7
rms $ f_{1001} / f_{1010} $ [10^{-4}]	49.3 / 41.5
median ε_x (nm) / ε_y (pm)	0.72 / 0.33
transverse tunes Q_x / Q_y	194.165 / 170.405
rms $\Delta D_{x/y}^{(2)}$ (mm)	1049 / 1563

FCC-ee LCC V106.2.3 @ Z
(updated error tolerances)

Assumed error tolerances **(updated)**

Section	Element	Misalignment			Rotation	Field error (rel.)
		σ_x (μm)	σ_y (μm)	σ_s (μm)	σ_θ (μrad)	
ARC	Dipole	1000	1000	500	1000	$k_0: 1 \times 10^{-3}$
	Quadrupole	50	50	100	50	$k_1: 2 \times 10^{-4}$
	Sextupole	50	50	100	50	$k_2: 2 \times 10^{-4}$
	Girder	150	150	500	150	–
STRAIGHT SECTION	IR Dipole	1000	1000	100	1000	$k_0: 1 \times 10^{-3}$
	TR Dipole	1000	1000	500	1000	$k_0: 1 \times 10^{-3}$
	FD Quadrupole	30	30	100	30	$k_1: 5 \times 10^{-5}$
	FF Quadrupole (excl. FD)	30	30	100	30	$k_1: 1 \times 10^{-4}$
	TR Quadrupole	100	100	100	100	$k_1: 2 \times 10^{-4}$
	FF Sextupole (incl. CS)	30	30	100	30	$k_2: 1 \times 10^{-4}$
I	BPM	10	10	–	100	–

Optics before correction

Observable	LCC Z-pole (V106.2.3)
rms orbit H / V (μm)	No closed orbit
rms $\Delta D_{x/y}$ (mm)	21.1 / 58.3
rms $\Delta \mu_{x/y}$ [10^{-4} 2π]	108 / 348
rms $\Delta \beta_{x/y} / \beta_{x/y}$ (%)	14.3 / 36.5
rms $ f_{1001} / f_{1010} $ [10^{-4}]	1724 / 331

Tuning results

Observable	LCC Z-pole (V106.2.3)
failed seeds	15 / 100
rms orbit H / V (μm)	137 / 138
rms H / V orbit corrector strength in ARC (μrad)	11.55 / 11.60
rms H / V orbit corrector strength in IR (μrad)	2.36 / 2.16
rms H / V orbit corrector strength in TR (μrad)	4.65 / 4.67
rms normal / skew optics corrector strength (mT)	0.85 / 0.53
rms $\Delta D_{x/y}$ (mm)	0.72 / 1.59
rms $\Delta\mu_{x/y}$ [10^{-4} 2π]	15.0 / 26.8
rms $\Delta\beta_{x/y}/\beta_{x/y}$ (%)	0.62 / 0.98
rms $ f_{1001} / f_{1010} $ [10^{-4}]	29.1 / 23.2
median ε_x (nm) / ε_y (pm) [from Twiss method]	0.71 / 0.038
transverse tunes Q_x/Q_y	194.16 / 170.20
rms $\Delta D_{x/y}^{(2)}$ (mm)	468 / 652

Corrected lattices with beam-beam @ 75 %

Observable	LCC Z-pole (V106.2.3)
number of lattices	82
rms $\Delta D_{x/y}$ (mm)	1.6 / 9.0
rms $\Delta \mu_{x/y}$ [10^{-4} 2π]	27.1 / 1216
rms $\Delta \beta_{x/y} / \beta_{x/y}$ (%)	2.12 / 42.7
rms $ f_{1001} / f_{1010} $ [10^{-4}]	49.3 / 41.5
median ε_x (nm) / ε_y (pm)	0.72 / 0.33
transverse tunes Q_x / Q_y	194.165 / 170.405
rms $\Delta D_{x/y}^{(2)}$ (mm)	1049 / 1563